

Lecture 13

What is quantum integrability?

Objectives. *Classical integrability was a definition with a theorem attached: n independent integrals in involution, and under two hypotheses the motion winds round tori. The quantum version has no such definition, and this lecture says why. Counting commuting operators decides nothing, because every Hamiltonian on a finite-dimensional space has as many as there are states. What physicists use instead is a set of criteria: conserved charges that are sums of local pieces, scattering without diffraction, and the statistics of the energy levels. The last can be tested on a laptop, and doing so honestly takes one piece of care that every first attempt leaves out. The lecture is the quantum mirror of [Lecture 6](#).*

13.1 What Part I hands over

Part II can be taken without Part I, and it rests on exactly six results from it. Each is restated in one line where it is used; here they are together.

- E1** *Brackets and first integrals* ([Lecture 1](#)). On phase space, $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$, observables evolve by $\dot{f} = \{f, H\}$, and f is a *first integral* when $\{f, H\} = 0$ ([Definition 1.13](#)).
- E2** *Liouville–Arnold* ([Theorem 4.8](#)). If $F_1, \dots, F_n$ are in involution, $dF_1, \dots, dF_n$ are linearly independent on a connected component Σ of a level set, and Σ is compact, then Σ is an n -torus on which the motion is linear in angle coordinates. Without compactness but with complete flows, the same proof gives $T^k \times \mathbb{R}^{n-k}$ ([§ 4.4](#)).
- E3** *Actions* ([Lecture 5](#)). $I_i = \frac{1}{2\pi} \oint_{\gamma_i} p \cdot dq$, and the old quantum theory’s rule $\oint p \cdot dq = 2\pi\hbar n$ ([\(5.18\)](#)).
- E4** *Lax pairs* ([Theorem 7.2](#)). If $\dot{L} = [M, L]$, every $\text{tr } L^k$ is conserved.
- E5** *The Sklyanin bracket* ([Theorem 8.17](#)). A monodromy matrix whose bracket is

$$\{T_1(\lambda), T_2(\mu)\} = [r_{12}(\lambda - \mu), T_1(\lambda)T_2(\mu)]$$

has transfer matrices in involution, $\{t(\lambda), t(\mu)\} = 0$ ([Corollary 8.18](#)). Separately, for the *linear* bracket $\{L_1, L_2\} = [r_{12}, L_1 + L_2]$ with a skew r , $r_{21} = -r_{12}$, the classical Yang–Baxter equation (CYBE) is sufficient for the Jacobi identity ([Theorem 8.11](#)).

- E6** *Scattering without diffraction* (Moser’s theorem, [Theorem 9.9](#), cited in [Lecture 9](#)). In the open Toda chain the set of outgoing momenta equals the set of incoming ones, and the net three-body shifts do not depend on the order of the collisions.

[Figure 13.1](#) lays out what Part II makes of each; it is the table that closed [Lecture 12](#), with the labels added.

	Part I	built in	becomes, in Part II
E1	$\{f, g\}, \dot{f} = \{f, H\}$, first integrals	Lecture 1	$[\cdot, \cdot] / i\hbar$; what counts as integrable (this lecture)
E2	Liouville–Arnold, with its hypotheses	Lecture 4	the question this lecture asks
E3	$\oint p \cdot dq = 2\pi\hbar n$	Lecture 5	Bethe equations, Lecture 15
E4	Lax pair, $\text{tr } L^k$ conserved	Lecture 7	the Lax operator $L_n(\lambda)$, Lecture 17
E5	Sklyanin bracket, $\{t(\lambda), t(\mu)\} = 0$, CYBE	Lectures 8 and 10	RTT , commuting transfer matrices, YBE, Lecture 16 , Lecture 17
E6	momenta permuted, shifts order-independent	Lecture 9	factorised scattering, Lecture 16
—	Calogero–Moser by reduction	Lectures 11 and 12	quantum Calogero–Sutherland, Lecture 21

Figure 13.1: Part I in one page: the table that closes [Lecture 12](#), with the export labels added. The six labelled rows are the only Part I results a Part II proof uses; the seventh is a comparison, not an import.

The rule of translation

Quantum mechanics keeps the shape of E1 and changes its ingredients. States are vectors in a Hilbert space, observables are Hermitian operators, and in the Heisenberg picture an observable evolves by

$$\frac{dA}{dt} = \frac{1}{i\hbar} [A, H]. \quad (13.1)$$

Set against $\dot{f} = \{f, H\}$, this is the substitution $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot] / i\hbar$ — checked on the one bracket that fixes it, $\{q, p\} = 1$ against $[q, p] = i\hbar$. An operator A is conserved exactly when $[A, H] = 0$, and two conserved operators are “in involution” when they commute. From here on $\hbar = 1$, except in [Lecture 17](#), where the classical limit $\hbar \rightarrow 0$ is taken and $\hbar$ is written.

So far every row translates. E2 is the one that does not, and the rest of this lecture is about why.

13.2 The naive definition fails

What “independent” meant classically

Every function of the energy is conserved, $\{h(H), H\} = 0$, so conserved quantities are never scarce. Liouville’s definition ([Definition 4.6](#)) asks for more: n of them whose *differentials* are linearly independent ([Definition 4.1](#)). That is a condition with content. $h(H)$ fails it, since $dh(H) = h'(H) dH$ is parallel to dH ; and independence is what makes a level set $\{F = f\}$ an n -dimensional manifold, which the theorem then proves to be a torus. At most n functions in involution can be independent ([Proposition 3.22](#)), so “ n ” is a sharp number.

The quantum count

Take the simplest quantum system that could be a many-body problem: a Hilbert space $\mathbb{C}^D$ of finite dimension. A chain of N spins $\frac{1}{2}$ has $D = 2^N$. Let H be Hermitian with *simple* spectrum $E_1 < E_2 < \dots < E_D$ and eigenvectors ψ_a , and let $\Pi_a = \psi_a \psi_a^\dagger$ be the projector onto the a -th eigenline. (The projectors are Π_a throughout; P is the permutation operator of [Definition 8.3](#).)

Proposition 13.1 (Every Hamiltonian has D commuting integrals). (a) The Π_a commute with H and with each other, are linearly independent, and $H = \sum_a E_a \Pi_a$.

(b) An operator A commutes with H if and only if $A = \sum_a c_a \Pi_a$ for some numbers c_a , if and only if $A = f(H)$ for a polynomial f of degree less than D . In particular

$$\Pi_a = \prod_{b \neq a} \frac{H - E_b}{E_a - E_b}. \quad (13.2)$$

(c) Every operator A on $\mathbb{C}^D$ satisfies a nonzero polynomial equation, $p_A(A) = 0$, where p_A is its characteristic polynomial.

Proof. (a) In the eigenbasis every Π_a and H are diagonal, so they commute; Π_a has a single 1 on the diagonal, in position a , so the D of them are linearly independent, and $H = \text{diag}(E_a) = \sum E_a \Pi_a$. (b) In the eigenbasis, $(AH - HA)_{ab} = (E_b - E_a)A_{ab}$, which vanishes for all $a \neq b$ exactly when A is diagonal, $A = \sum_a A_{aa} \Pi_a$. The right side of (13.2) is diagonal with entry $\prod_{b \neq a} (E_c - E_b)/(E_a - E_b)$ in position c , which is 1 for $c = a$ and 0 otherwise; so each Π_a , hence each $\sum c_a \Pi_a$, is a polynomial in H of degree $D - 1$. Conversely every polynomial in H commutes with H . (c) is the Cayley–Hamilton theorem. ■

Now try to say what “ D independent conserved operators” should mean.

- *Linearly independent.* By (a), every Hamiltonian with simple spectrum has D of them in involution, $D - 1$ after discarding the identity $\sum \Pi_a$ (with degenerate levels, take the projectors of any eigenbasis). A random matrix qualifies as well as the XXX chain of § 13.3.
- *Functionally independent,* the classical notion, where $A = f(B)$ counts as dependent, as $h(H)$ did. By (b), if the spectrum is simple then every conserved operator is a function of H , and no such Hamiltonian has a single integral beyond H itself, however regular its dynamics. With degenerate levels, as for XXX, the same happens one step up: every commuting family consists of functions of one operator (von Neumann; the reading at the end of this lecture).
- *Algebraically independent:* no nonzero polynomial relation among them. By (c), not even one operator on $\mathbb{C}^D$ is algebraically independent; the projectors satisfy $\Pi_a^2 = \Pi_a$ and $\Pi_a \Pi_b = 0$ besides.

One reading makes every system integrable, the other two make none. The count of commuting operators cannot be the definition, and there is no quantum analogue of functional independence to rescue it: the classical notion was a statement about differentials, and a finite matrix has none. This is the remark of § 0.3 made precise; Caux and Mossel (2011) call the counting definition “too naive” for this reason · *cited*.

What survives is not *how many* conserved operators there are but *what they look like* in the basis in which the system was written down. The projector onto a typical eigenstate of a chain of twenty spins has nonzero entries all over its symmetry block, a block of dimension up to about 10^4 ; the Hamiltonian is a sum of twenty terms, each acting on two neighbouring spins. That difference is the next section.

13.3 Local and extensive charges

Densities

From here on the system is a ring of N sites, each carrying a spin $\frac{1}{2}$: the Hilbert space is $(\mathbb{C}^2)^{\otimes N}$, site $N+1$ is site 1, and τ is the translation that moves every site one step, $n \mapsto n+1$. An operator *acts on* a set of sites if it is the identity on every other factor.

Definition 13.2 (Local charge). A density of range r is an operator q acting on sites $1, \dots, r$. On a ring of $N \geq r$ sites its *local charge* is the sum of its translates,

$$Q = \sum_{n=1}^N \tau^{n-1}(q),$$

$\tau^{n-1}(q)$ acting on sites $n, \dots, n+r-1$. The range r and the density q are fixed once and for all, and N varies: a local charge is really one operator for each chain length. A local charge that commutes with H is a *local conserved charge*.

The fixed range is the whole content: with $r = N$ allowed, any translation-invariant operator A would qualify, as the sum of the translates of A/N . The total spin $S^z = \sum_n S_n^z$ is a local charge of range 1, and so is H of (13.3) below, of range 2. Local charges are *extensive*: Q is a sum of N terms of norm $\|q\|$, so $\|Q\| \leq N \|q\|$, and in a translation-invariant state $\langle Q \rangle = N \langle q \rangle$ — proportional to the length, as energy and particle number are. And local charges are rare.

Proposition 13.3 (Local charges are few). For each fixed r , the local charges of range at most r on a ring of $N \geq r$ sites form a vector space of dimension at most 4^r , whatever N . So among the 2^N linearly independent conserved operators of a Hamiltonian with simple spectrum, the local conserved charges of range at most r span at most 4^r dimensions.

Proof. A density of range r is an operator on $(\mathbb{C}^2)^{\otimes r}$, a space of dimension 4^r , and $q \mapsto Q$ is linear. A density of smaller range is one of range r that acts trivially on the last sites. ■

The bound does not grow with N , while the commutant has 2^N dimensions: at any fixed range, the local charges fill a fraction at most $4^r/2^N$ of the conserved operators, which goes to zero as the chain grows. So no family of local charges of one bounded range can be complete; a tower whose ranges grow, like XXX's below, needs a separate count, and on a ring of N sites it has only about N members. Locality — not number — is what can tell one Hamiltonian's conserved quantities from another's. It is the preferred structure Caux and Mossel ask for.

The XXX chain and its third charge

The spin- $\frac{1}{2}$ Heisenberg chain is

$$H = J \sum_{n=1}^N \left(\mathbf{S}_n \cdot \mathbf{S}_{n+1} - \frac{1}{4} \right) = \frac{J}{2} \sum_{n=1}^N (P_{n,n+1} - 1), \quad J > 0, \quad (13.3)$$

with $\mathbf{S} = \boldsymbol{\sigma}/2$ and $P_{n,n+1}$ the permutation of the two neighbouring factors (Definition 8.3). The second form uses $\boldsymbol{\sigma}_n \cdot \boldsymbol{\sigma}_{n+1} = 2P_{n,n+1} - 1$, true because both sides equal 1 on the triplet and

−3 on the singlet. Write $P_n := P_{n,n+1}$, a density of range 2. The only facts about permutations the next proof uses are

$$P_n^2 = 1, \quad P_n P_{n+1} P_n = P_{n+1} P_n P_{n+1}, \quad P_n P_m = P_m P_n \text{ if } |n - m| \geq 2, \quad (13.4)$$

the last because the two bonds share no site (on the ring, $|n - m|$ is measured round it). The middle identity says both sides exchange sites n and $n + 2$.

Theorem 13.4 (A third conserved charge). *On a ring of $N \geq 4$ sites, the local charge of range 3*

$$Q_3 = \sum_{n=1}^N (\sigma_n \times \sigma_{n+1}) \cdot \sigma_{n+2} = 2i \sum_{n=1}^N [P_n, P_{n+1}] \quad (13.5)$$

commutes with the Heisenberg Hamiltonian (13.3).

Proof. The two forms agree. $[\sigma^a, \sigma^b] = 2i\epsilon_{abc}\sigma^c$ on one site gives $[\sigma_n \cdot \sigma_{n+1}, \sigma_{n+1} \cdot \sigma_{n+2}] = -2i(\sigma_n \times \sigma_{n+1}) \cdot \sigma_{n+2}$, and the left side is $4[P_n, P_{n+1}]$.

Only neighbours contribute. By (13.3), $[H, Q_3] = iJ \sum_{m,n} [P_m, [P_n, P_{n+1}]]$. The bond m commutes with P_n and P_{n+1} unless it touches one of the sites $n, n + 1, n + 2$, that is unless $m \in \{n - 1, n, n + 1, n + 2\}$ (four distinct bonds, since $N \geq 4$). The sum splits into two kinds of term (Figure 13.2).

Three-site terms, $m = n$ and $m = n + 1$. Write $a = P_n, b = P_{n+1}$. Using $a^2 = b^2 = 1$,

$$[a, [a, b]] = 2b - 2aba, \quad [b, [a, b]] = 2bab - 2a,$$

and by the braid relation $aba = bab$ the sum is $2(b - a) = 2(P_{n+1} - P_n)$. Summed over n round the ring, it telescopes to zero.

Four-site terms, $m = n - 1$ and $m = n + 2$. Relabelling $n \rightarrow n + 1$ in the first,

$$\sum_n \left([P_{n-1}, [P_n, P_{n+1}]] + [P_{n+2}, [P_n, P_{n+1}]] \right) = \sum_n \left([a, [b, c]] + [c, [a, b]] \right),$$

now with $a = P_n, b = P_{n+1}, c = P_{n+2}$. The Jacobi identity says $[a, [b, c]] + [c, [a, b]] = -[b, [c, a]]$, and $[c, a] = 0$ because the bonds n and $n + 2$ are disjoint. So $[H, Q_3] = 0$. ■

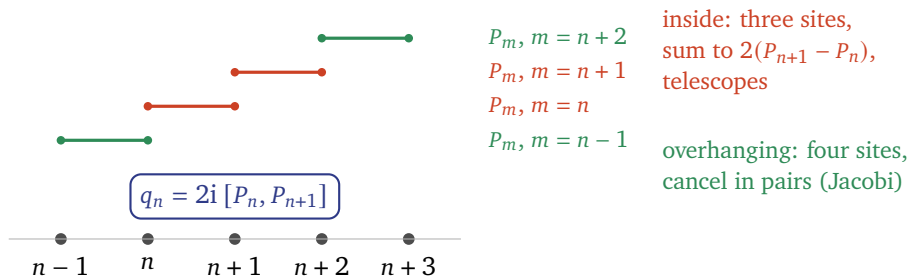


Figure 13.2: The stencil of $[H, Q_3]$. The density q_n of Q_3 spans three sites; four bonds of H touch it. The two inside it give three-site terms that add up to a difference of neighbouring bonds; the two overhanging it give four-site terms that cancel against the neighbouring densities' by the Jacobi identity. Every other bond commutes with q_n .

Q_3 is genuinely new: it is not a polynomial in H (the tutorial shows this by a reflection). And the proof is fragile. Add a next-nearest-neighbour coupling $J_2 \sum \mathbf{S}_n \cdot \mathbf{S}_{n+2}$ — the

J_1 – J_2 chain — and both cancellations fail, and a computer search at $N = 8, 9$ finds no range-3 conserved charge beyond H and the total spin (the tutorial). Make the chain anisotropic instead (XXZ) and (13.5) fails too, but a modified range-3 charge exists: XXZ is integrable. Grabowski and Mathieu proposed the existence of one range-3 charge beyond the Hamiltonian as a test of integrability; it is a conjecture · *cited*.

The tower. XXX has a whole tower of commuting local charges, $Q_2 \propto H$, Q_3 , Q_4 , ..., with Q_k of range k (Grabowski and Mathieu 1994) · *cited*. Lecture 17 obtains them all from one commuting family of transfer matrices $t(\lambda)$, as the classical $\text{tr } L^k$ came from one Lax matrix. On a ring of N sites there are about N of them: a handful beside 2^N projectors. Beyond them lie *quasi-local* charges, whose densities decay exponentially with their range instead of stopping (Ilievski, Medenjak, Prosen and Zadnik 2016) · *cited*; Lecture 23 needs them.

13.4 Level statistics

Conserved charges are structure one has to find. The spectrum is something one can simply compute, and it carries a signature of integrability.

Spacings

Let $E_1 \leq E_2 \leq \dots$ be the eigenvalues of H in one symmetry sector, a notion made precise below, and $s_n = E_{n+1} - E_n$ the spacings. The density of levels varies across the spectrum, so before spacings from different places can be compared they are *unfolded*: replace each E_n by the smoothed count of levels below it, so that the mean spacing is 1 everywhere. The histogram $P(s)$ of unfolded spacings is the object.

Uncorrelated levels. If the levels were independent random points of unit density, the chance that an interval of length s contains none of them would be e^{-s} (the limit of $(1 - s/L)^L$ for L points in length L), and the spacing distribution is

$$P_P(s) = e^{-s} \quad (\text{Poisson}). \quad (13.6)$$

Its maximum is at $s = 0$: nearly degenerate pairs are common.

Levels that repel. Now let the Hamiltonian be a generic real symmetric matrix, with no structure. For two levels the whole story is visible.

Proposition 13.5 (The Wigner surmise). Let $H = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ with a, b Gaussian of variance 1 and c Gaussian of variance $\frac{1}{2}$, all independent (the Gaussian orthogonal ensemble, GOE, for 2×2). Its eigenvalue spacing, rescaled to mean 1, has density

$$P_W(s) = \frac{\pi s}{2} e^{-\pi s^2/4}. \quad (13.7)$$

Proof. The spacing is $s = \sqrt{x^2 + y^2}$ with $x = a - b$ and $y = 2c$, which are independent Gaussians of the same variance, 2. So (x, y) is an isotropic Gaussian in the plane and s its distance from the origin: its density is proportional to the circumference $2\pi s$ times $e^{-s^2/4}$. Rescaling to mean 1 gives (13.7). ■

The factor s is the whole point. A degeneracy needs *two* conditions, $a = b$ and $c = 0$, and two conditions on a random point of the plane are met with probability zero and nearly met only rarely: *level repulsion*, $P(s) \rightarrow 0$ as $s \rightarrow 0$. (For complex Hermitian matrices

the off-diagonal entry has two real parts, three conditions, and $P \propto s^2$ near 0.) For large GOE matrices the spacing distribution is not exactly (13.7), but close to it: pooling $3 \cdot 10^4$ bulk spacings of 100 random 600×600 GOE matrices, the fraction below any s differs from the surmise's by 0.007 (check-113-algebra.py, which requires less than 0.015). “Wigner–Dyson statistics” means this shape.

Why should an integrable system be Poissonian and a chaotic one look like a random matrix? Two conjectures say so, supported by much evidence and proved in neither generality: Berry and Tabor (1977) for integrable systems and Bohigas, Giannoni and Schmit (1984) for chaotic ones · *cited*; spin chains were tested from Poilblanc et al. (1993) on. The picture to keep is the trap below in disguise. In an integrable system every level carries quantum numbers, one per conserved charge; levels with different labels do not mix, so their spacings are uncorrelated. In a chaotic system nothing labels the levels, and every pair repels. (The heuristic behind the conjectures is in the reading at the end of this lecture.)

The ratio

Unfolding needs a fit and a choice of what to discard at the edges. A statistic that needs neither is the ratio of consecutive spacings (Oganesyan and Huse 2007),

$$r_n = \frac{\min(s_n, s_{n+1})}{\max(s_n, s_{n+1})} \in [0, 1], \quad (13.8)$$

which does not see the local density because it is a ratio. For Poisson levels its mean is

$$\langle r \rangle_P = 2 \ln 2 - 1 \approx 0.386 \quad (13.9)$$

(the tutorial at the end of this lecture); for the GOE, $\langle r \rangle \approx 0.531$ (Atas, Bogomolny, Giraud and Roux 2013, who find 0.5307 numerically for large matrices and $4 - 2\sqrt{3} \approx 0.536$ from 3×3 ones) · *cited*. One number, between 0.386 and 0.531, is the whole diagnosis.

The trap: resolve the symmetries first

Here is the first experiment everyone does, and it gives the wrong answer. Take a chain that nobody believes to be integrable, compute its levels, and histogram the spacings (Figure 13.3, left). The histogram is Poisson.

The reason is block structure. If H commutes with a symmetry — here the translation τ — it is block diagonal in the symmetry's eigenbasis, one block per eigenvalue (per momentum k). Levels in different blocks sit in different blocks of a matrix: they never mix, so they do not repel, and a list that interleaves several blocks' spectra is a superposition of independent sequences. Superpose enough independent sequences and the result is Poisson whatever each one was — the same reason independent points are Poisson. (Two GOE spectra superposed already give $\langle r \rangle \approx 0.43$; eight give 0.39; check-113-algebra.py.) So a *symmetry left unresolved fakes integrability*. The cure is to diagonalise one block at a time and collect statistics inside each.

Which symmetries. Here: translation (momentum $k = 2\pi j/N$), the total S^z , the spin flip $S^z \rightarrow -S^z$, the reflection $k \rightarrow -k$, and for isotropic chains $SU(2)$. A block is *resolved* when no symmetry acts within it; $S^z = 1$, $0 < k < \pi$ and anisotropy achieve it, since the flip and the reflection each move such a block to a different one. That argument is the proof. The generator adds a necessary check, not a sufficient one: no two levels in any block coincide (no spacing below 10^{-8} of the mean). A reader repeating the experiment should do both.

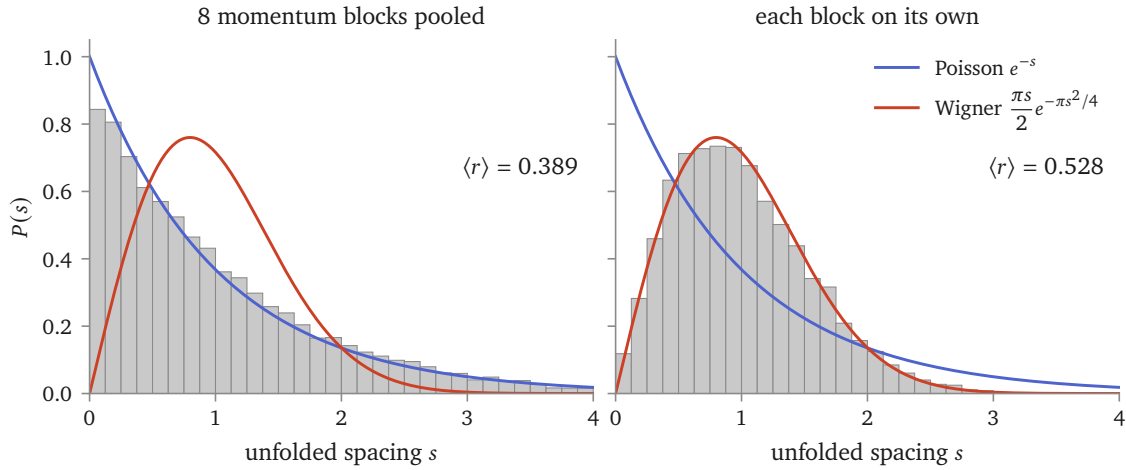


Figure 13.3: One integrability-breaking chain, (13.10) with $\Delta = 0.55$, $J_2 = 0.5$, $N = 18$, $S^z = 1$. Left: the levels of the eight momentum blocks $0 < k < \pi$ put into one list — Poisson, and wrong. Right: spacings taken inside each block and then pooled — Wigner–Dyson. The eigenvalues are the same in both panels. (Generated and checked by `scripts/figures/level_statistics.py`.)

The experiment

The chain is the XXZ chain with a next-nearest-neighbour coupling,

$$H = \sum_{n=1}^N (S_n^x S_{n+1}^x + S_n^y S_{n+1}^y + \Delta S_n^z S_{n+1}^z) + J_2 \sum_{n=1}^N (S_n^x S_{n+2}^x + S_n^y S_{n+2}^y + \Delta S_n^z S_{n+2}^z). \quad (13.10)$$

At $J_2 = 0$ it is the XXZ chain, which [Lecture 18](#) solves by the algebraic Bethe ansatz; for $J_2 \neq 0$ it is not known to be integrable, and [Figure 13.4](#) shows why nobody expects it to be. The anisotropy is $\Delta = 0.55$, away from the values $\cos(\pi p/q)$ with small q at which XXZ acquires extra degeneracies.

[Figure 13.5](#) sweeps J_2 . The mean ratio leaves the Poisson value as soon as J_2 is switched on and reaches the GOE value by $J_2 \approx 0.4$; the crossover is sharper for the longer chain. Whether an arbitrarily small J_2 makes an infinitely long chain chaotic is a real question, and the figure cannot answer it; [§ 13.6](#) returns to it.

13.5 Working criteria, and their status

The working criteria of quantum integrability, as used in practice and in the rest of this course:

- (T) *Transfer-matrix charges*. A family of commuting transfer matrices $t(\lambda)$, built from a solution of the Yang–Baxter equation, whose logarithmic derivatives give a tower of local conserved charges ([Lecture 17](#)).
- (S) *Non-diffractive scattering*. When particles are far apart, their momenta are conserved as a set, and many-body scattering factorises into two-body events — the quantum form of E6, and Sutherland’s definition ([Lecture 15](#), [Lecture 16](#)).
- (P) *Poisson statistics*, in fully resolved symmetry blocks.

They are criteria, not a definition, because no theorem connects them, and there are systems where one holds without another.

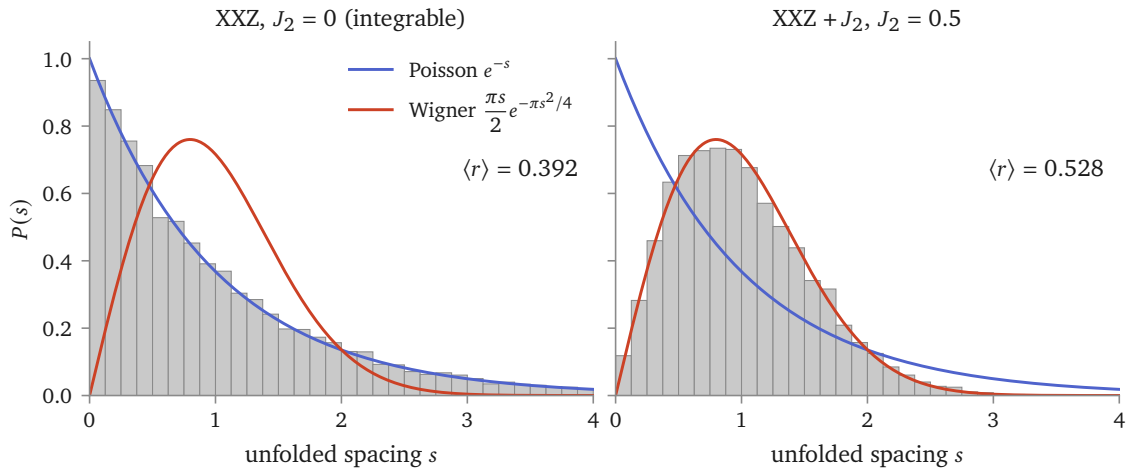


Figure 13.4: Unfolded spacings of (13.10), $N = 18$, $S^z = 1$, each momentum block $0 < k < \pi$ unfolded on its own and the spacings pooled (8 blocks, 19 448 levels, the central 80% of each used). Left: XXZ, integrable, Poisson. Right: with $J_2 = 0.5$, Wigner–Dyson. The generator asserts $|\langle r \rangle - 0.386| < 0.015$ on the left and $|\langle r \rangle - 0.531| < 0.015$ on the right.

- *Charges without (P).* The Haldane–Shastry chain, whose exchange falls off as the inverse square of the chord distance, has an explicit commuting family of conserved charges (Talstra and Haldane 1995), though not local ones: they are sums over pairs, triples, ... of sites. Its level statistics are neither Poisson nor Wigner–Dyson (Finkel and González-López 2005, quoted by Caux and Mossel) · *cited*. The harmonic oscillator, the most integrable system there is, is not Poisson either: equally spaced levels in one dimension, highly degenerate ones with commensurate frequencies (the reading at the end of this lecture).
- *(P) without (T).* A strongly disordered chain in the many-body localised regime shows Poisson statistics and has no transfer matrix; it has quasi-local conserved quantities of its own, but whether that regime survives in infinitely long chains is contested (Abanin, Altman, Bloch and Serbyn 2019; Šuntajs, Bonča, Prosen and Vidmar 2020) · *cited*. And, as Figure 13.3 showed, (P) can be produced by carelessness alone.
- *(S) needs room.* Non-diffraction concerns particles that separate — gases on a line, dilute excitations of a chain — and says nothing directly about a finite spectrum (Caux and Mossel) · *cited*.

Symmetry is not integrability, in either direction. The J_1 – J_2 Heisenberg chain has all of $SU(2)$ and translation, so its levels come in large degenerate multiplets; yet within blocks of fixed total spin and momentum its statistics are Wigner–Dyson ($\langle r \rangle = 0.53$ at $J_2 = 0.7$, $N = 18$, total spin 1; check-113-algebra.py), and a search at that coupling ($N = 8, 9$) finds no range-3 charge beyond H and the total spin (§ 13.3). Conversely the anisotropic XYZ chain, $\sum (J_x S^x S^x + J_y S^y S^y + J_z S^z S^z)$ with three different couplings, has no continuous symmetry at all and is integrable, by a commuting family of transfer matrices (Baxter, *Exactly Solved Models*, §10.14) · *cited*. Degeneracies count symmetries; integrability is something else.

What this course does. Every model of Part II that is called integrable is known to have (T) — constructed in full for XXX in Lecture 17, which also verifies quantum Toda’s *RLL* relation and its classical limit, and cited for the others — or has its charges constructed directly (Lecture 14, Lecture 21). Each lecture names the criterion it uses; none is assumed

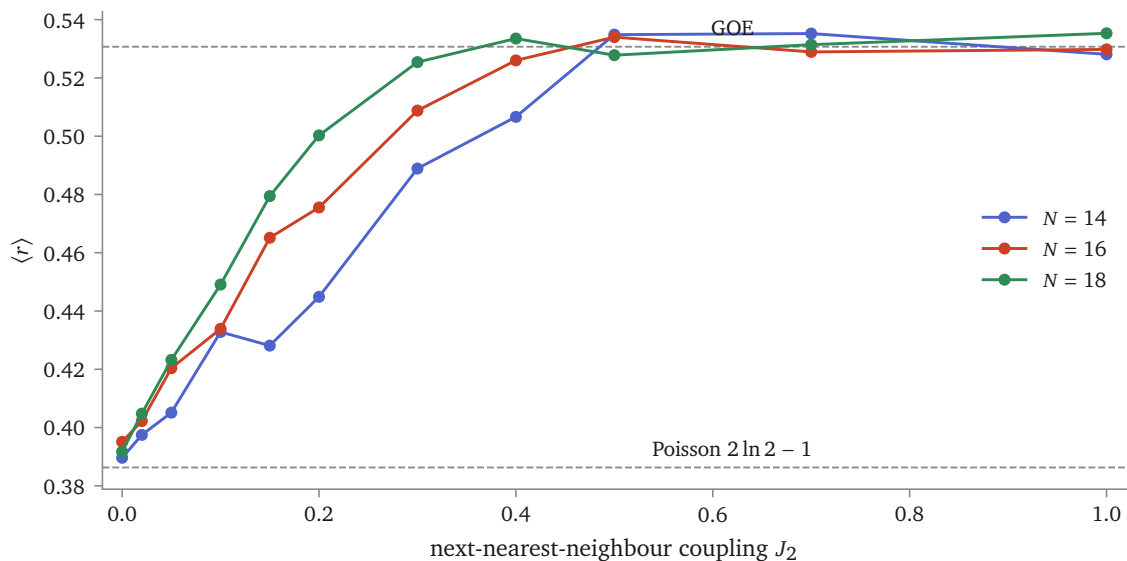


Figure 13.5: $\langle r \rangle$ against the integrability-breaking coupling J_2 in (13.10), resolved blocks, three chain lengths. The crossover from Poisson to GOE moves to smaller J_2 as N grows.

to imply another.

13.6 The mirror of Lecture 6

Lecture 6 diagnosed classical integrability by a picture: curves or scattered dots in a Poincaré section (Figure 6.9). Figure 13.6 sets it beside this lecture's. No theorem relates the two rows — one is a property of orbits, the other of a spectrum — but they play the same role, and the two conjectures above claim that they agree where both make sense.

The mirror goes one step further. Classically, a weak perturbation of an integrable system does not destroy it at once: most tori survive (KAM, Theorem 6.8), and chaos grows from the resonances. The quantum question is what Figure 13.5 raises and cannot settle: does a weakly broken integrable chain keep a memory of its charges? In real time it often does, for a while: for suitable initial states, and when the breaking is weak enough to separate the time scales, a system started far from equilibrium first settles to a state set by the approximate charges, and thermalises only much later (Mori, Ikeda, Kaminishi and Ueda 2018) · cited. That *prethermalisation* is the nearest quantum relative of KAM — an analogy, not a theorem — and it is Lecture 23's.

Dictionary

the Poisson bracket, $\dot{f} = \{f, H\} \rightarrow$ the commutator, $\dot{A} = [A, H] / i\hbar$ (analogy: the same role; this lecture proves nothing about $\hbar \rightarrow 0$)

Liouville's n independent integrals in involution $\rightarrow$ no agreed definition; three criteria, none implying another (a declared absence)

Poincaré sections: curves or scattered dots $\rightarrow$ level statistics: Poisson or Wigner–Dyson (analogy; the link is the Berry–Tabor and BGS conjectures, cited)

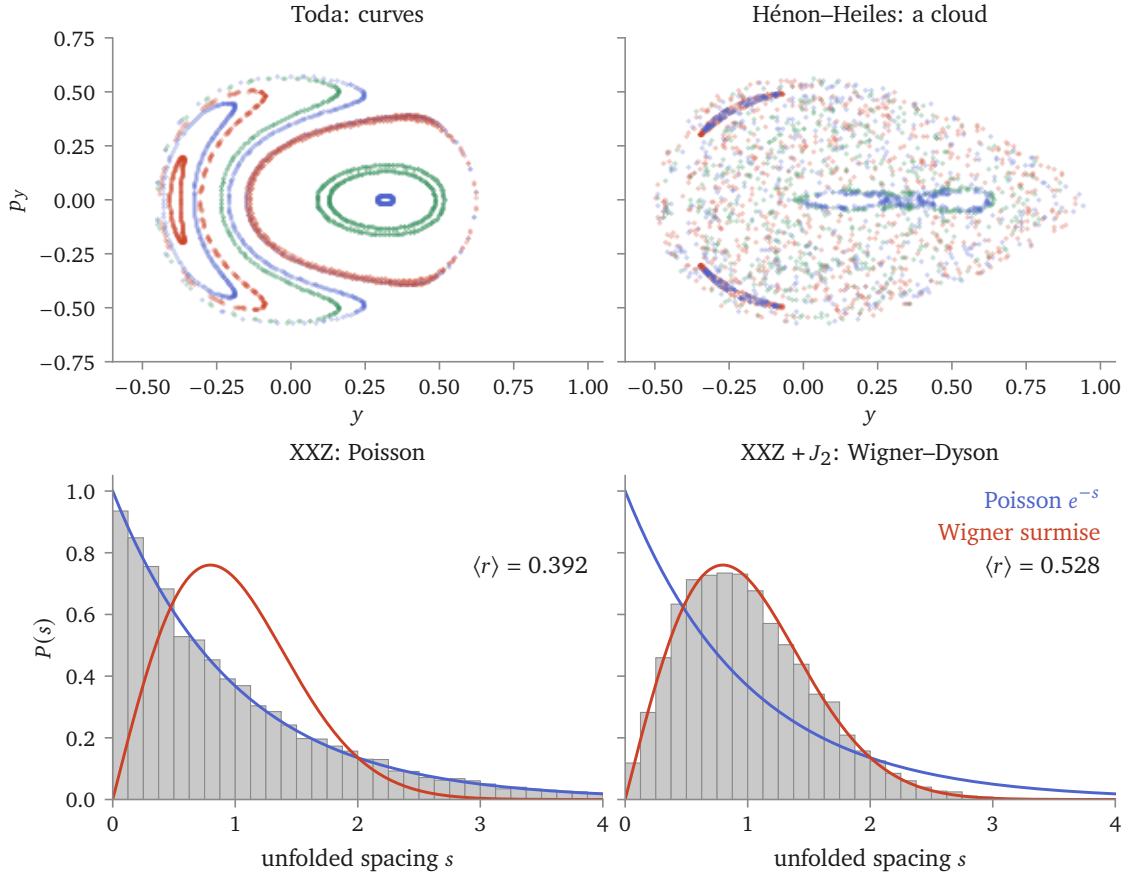


Figure 13.6: The two diagnostics. Top (Lecture 6, Figure 6.9): Poincaré sections at $E = 1/6$ of the three-particle Toda chain and of its cubic truncation, Hénon-Heiles. Bottom (Figure 13.4): spacings of XXZ and of XXZ with $J_2 = 0.5$. Curves against a cloud (with the few islands of regular motion Lecture 6 found); Poisson against Wigner-Dyson. (scripts/figures/quantum_mirror.py.)

Tutorial

13.7 Two computations

Q_3 is not a polynomial in H . Let R be the reflection of the ring, site $n \mapsto N + 1 - n$. It maps each bond to a bond, so $RHR^{-1} = H$, and hence $Rf(H)R^{-1} = f(H)$ for every polynomial f . It maps $(\sigma_n \times \sigma_{n+1}) \cdot \sigma_{n+2}$ to the same triple product of three consecutive sites taken in the reverse order, which is minus it (the three factors act on different sites and commute, and the triple product is antisymmetric in its outer two vectors). So $RQ_3R^{-1} = -Q_3$. Finally $Q_3 \neq 0$ for $N \geq 4$: its terms are Pauli strings on different triples of sites and cannot cancel.

A search for local charges. Every local charge of range at most 3 is a combination of the translation sums of the $4^3 = 64$ Pauli strings on three sites; for $N \geq 5$ they span 49 dimensions (Proposition 13.3; on smaller rings wrap-around makes some coincide). Requiring $[H, Q] = 0$ is a linear condition on the coefficients, solved by computer. At $N = 8$ and 9 (scripts/check-113-algebra.py) the solutions are: for XXX exactly $1, S^x, S^y, S^z, H$ and Q_3 ; for the J_1 - J_2 chain at $J_2 = 0.3, 0.5$ and 0.7 the same list without Q_3 ; for XXZ at $\Delta = 0.55, 1, S^z, H$ and an anisotropic cousin of Q_3 ; for XXZ with J_2 , only $1, S^z, H$. That is a search at two sizes, not a proof about any other.

The ratio for uncorrelated levels. Take two consecutive spacings of Poisson levels, independent with density e^{-s} . Then $\tilde{r} = s_n/s_{n+1}$ has distribution

$$\Pr(\tilde{r} \leq x) = \int_0^\infty e^{-t} (1 - e^{-xt}) dt = 1 - \frac{1}{1+x}, \quad \text{density } \frac{1}{(1+x)^2}.$$

The density is unchanged by $x \mapsto 1/x$ (with its Jacobian), so $r = \min(\tilde{r}, 1/\tilde{r})$ has density $2/(1+r)^2$ on $[0, 1]$, and

$$\langle r \rangle_P = \int_0^1 \frac{2r}{(1+r)^2} dr = 2 \left[\ln(1+r) + \frac{1}{1+r} \right]_0^1 = 2 \ln 2 - 1.$$

The same computation with the three-level GOE density in place of the exponentials gives $4 - 2\sqrt{3}$ (Atas et al.). The check script confirms both, and confirms by simulation that the ratio of two independent exponentials has the density above.

Reading

13.8 Three refinements

Degenerate spectra. If H has degenerate eigenvalues — because of a symmetry, say — the operators commuting with it no longer commute with each other, and one looks instead for a maximal commuting family among them. Any such family, simultaneously diagonalised, still consists of the linear combinations of D projectors, and the objections of § 13.2 return. Von Neumann showed that any commuting family of Hermitian matrices consists of functions of a single one of them (cited by Caux and Mossel) · *cited*.

Which ensemble. The momentum blocks of § 13.4 are complex Hermitian, which would suggest the s^2 repulsion of the unitary ensemble. But reflection combined with complex conjugation maps each block to itself, and that antiunitary symmetry, squaring to +1, makes the real, orthogonal ensemble the right reference — which is what Figure 13.4 shows. In general $P(s)$ vanishes like s , s^2 or s^4 — Dyson’s index 1, 2 or 4 — for the orthogonal (an antiunitary symmetry squaring to +1), unitary (none) and symplectic (one squaring to −1) ensembles. Lecture 21 meets all three again, in the ground states of the Calogero–Sutherland model.

Classes. Caux and Mossel propose a classification by how fast the number of nonzero coefficients of the charges grows with N when they are expanded in a preferred tensor-product basis of operators, such as the Pauli strings ($\sum_n S_n^z$ has N of them, although as a matrix it has exponentially many nonzero entries): XXX is “linear”, Haldane–Shastry “quasi-polynomial”, and a generic chain’s conserved operators beyond 1, H and its symmetries are expected to need exponentially many · *cited*. It is a proposal; this course uses only its moral, which is Proposition 13.3’s.

13.9 Why Poisson, and why Wigner–Dyson

Berry and Tabor. For an integrable system with n degrees of freedom semiclassical quantisation puts the levels at $E = K(I)$ with the actions on a lattice, $I = \hbar(m + \alpha)$, $m \in \mathbb{Z}^n$ (E3, with the Maslov shifts α of [Lecture 15](#)). The levels in a window of energy are the lattice points in a thin shell between two surfaces $K = E$ and $K = E + \delta E$; for $n \geq 2$ and a curved, generic K , successive lattice points met in energy order come from unrelated parts of the lattice, and Berry and Tabor (1977) argued that their spacings are then Poisson distributed · *cited*. The argument fails where the lattice is seen coherently. For harmonic oscillators K is linear in I : with one degree of freedom the levels are equally spaced (a “picket fence”); with commensurate frequencies they are highly degenerate; with incommensurate ones they are neither, and their spacing law depends on the arithmetic of the frequency ratio — but it is not Poisson either (Berry and Tabor treat both cases). The conjecture is for generic integrable systems.

Bohigas, Giannoni and Schmit. For a system whose classical limit is chaotic, the conjecture (1984) is that the spectral fluctuations are those of the random-matrix ensemble with the same antiunitary symmetry: orthogonal with a time reversal squaring to +1, unitary without one, symplectic with one squaring to -1 · *cited*. Semiclassical arguments through periodic orbits support it; there is no proof.

Spin chains have no classical limit with N fixed, so neither conjecture applies to them as stated; that they nevertheless follow the same pattern is an empirical fact, tested since the early 1990s and on every chain studied in this course, and it is why (P) is a criterion and not a theorem.

Summary.

- *The translation:* $\{\cdot, \cdot\} \rightarrow [\cdot, \cdot] / i\hbar$; A is conserved iff $[A, H] = 0$ ((13.1)).
- *Counting fails:* on $\mathbb{C}^D$, a Hamiltonian with simple spectrum has D linearly independent commuting conserved operators, all of them polynomials in H , and no operator is algebraically independent ([Proposition 13.1](#)). No reading of “independent” separates integrable from not.
- *Locality selects:* local charges of range $\leq r$ span at most 4^r dimensions, whatever N ([Proposition 13.3](#)).
- *XXX has a third charge:* $Q_3 = \sum (\sigma_n \times \sigma_{n+1}) \cdot \sigma_{n+2}$ commutes with H , by a telescoping argument using only $P^2 = 1$, the braid relation and Jacobi ([Theorem 13.4](#)). At $N = 8, 9$ a search finds no range-3 charge for the J_1 – J_2 chain beyond H and the total spin; XXZ has an anisotropic successor to Q_3 .
- *Level statistics:* Poisson e^{-s} against Wigner $\frac{\pi s}{2} e^{-\pi s^2/4}$ ([Proposition 13.5](#)); $\langle r \rangle \approx 0.386$ against 0.531; *resolve every symmetry first*, or Poisson is faked ([Figure 13.3](#)).
- *Criteria, not a theorem:* transfer-matrix charges, non-diffractive scattering, Poisson statistics; each is known to hold without another somewhere.

Tutorial. Exercise Sheet 13.

References

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